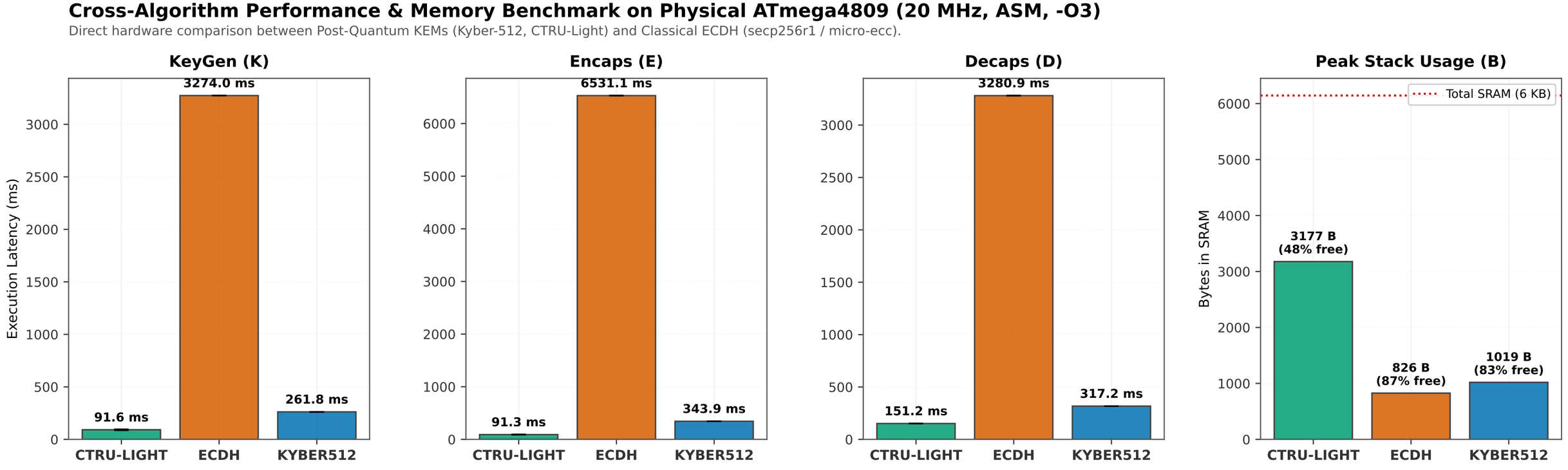
Cross-Algorithm Performance & Memory Benchmark on Physical ATmega4809 (20 MHz, ASM, -O3)
Direct hardware comparison between Post-Quantum KEMs (Kyber-512, CTRU-Light) and Classical ECDH (secp256r1 / micro-ecc).
KeyGen (K)
Execution Latency (ms)
91.6 ms
3274.0 ms
261.8 ms
Encaps (E)
91.3 ms
6531.1 ms
343.9 ms
Decaps (D)
151.2 ms
3280.9 ms
317.2 ms
CTRU-LIGHT
ECDH
KYBER512
Peak Stack Usage (B)
Bytes in SRAM
Total SRAM (6 KB)
3177 B
(48% free)
826 B
(87% free)
1019 B
(83% free)